TANGGAL PERCOBAAN :

TANGGAL PENGUMPULAN :

PRAKTIKUM PEMROGRAMAN KOMPUTER SEMESTER 125

MODUL – 5 Mengurutkan (Sorting)

Mencerdaskan & Memartabatkan Bangsa

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Laporan Awal	Laporan Akhir	Total
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Modul 6

Mengurutkan (Sorting)

1. Problem Statement
   = Diberikan 100 data nilai ujian yang tersimpan dalam sebuah list dan belum terurut.

2. Mathematical Equation
   = a. Himpunan data
   $x \{x_1, x_2, x_3, \ldots, x_n\}$ dengan $n = 100$

   b. Tujuan pengurutan
   hasil akhir $x' = \{x'_1, x'_2, \ldots, x'_n$ harus memenuhi:
   $x'_1 \leq x'_2 \leq x'_3 \leq \ldots \leq x'_n$

   c. Jumlah perbandingan
   $\sum (n-i)$ untuk $i = 1$ sampai $n-1 = n(n-1)/2 = 100 \times 99 / 2 = 4950$ perbandingan

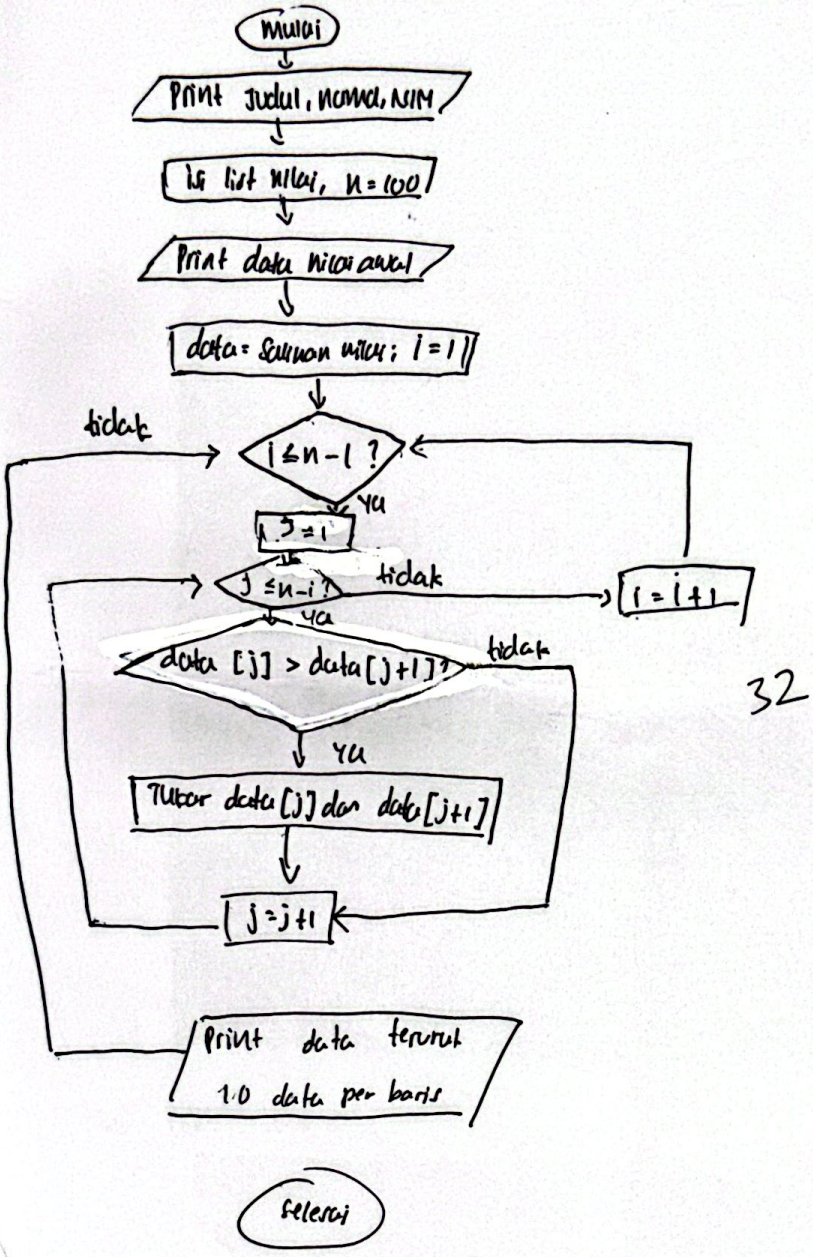
3. Algorithm
   1. Mulai
   2. Print "program mengurutkan (Sorting)"
   3. Print "Nama : Dwihayu khaila putri"
   4. Print " NIM : 1306625044"
   5. Deklarasikan list nilai berisi 100 data
   6. n = 100
   7. print "data Nilai Awal :" lalu seluruh list nilai
   8. data = salinan dari nilai
   9. i = 1
   10. selama i ≤ n - 1 kerjakan:
   11. j = 1
   12. selama j ≤ n-i kerjakan:
   13. jika data [j] > data [j+1] maka
   14. temp ← data [j]
   15. data [j] ← data [j+1]
   16. data [j+1] ← temp
   17. j ← j+1
   18. i ← i+1
   19. Print "data nilai terurut:"
   20. k ← 1
   21. selama k ≤ n kerjakan:
   22. cetak data [k]
   23. jika k mod 10 = 0 maka pindah baris

24. k ← k+1

25. Print "selesai

26. Selesai

4. Flowchart



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5.) Print out SC

SC-MODUL-5-1306625044-DKP.py

```python
# Program Sorting

garis1 = "=" * 50
garis2 = "-" * 50

# 1. Cetak judul, nama, dan NRM
print(garis1)
print("PROGRAM SORTING".center(50))
print(garis1)
print("Nama : Dyahayu Khaila Putri")
print("NIM  : 1306625044")
print(garis2)

# 2. Data nilai
nilai = [85, 13, 99, 34, 71, 15, 82, 24, 64, 61, 67, 99, 50, 68, 25, 37, 32, 27, 14, 91,
         79, 15, 47, 48, 74, 88, 64, 53, 77, 50, 24, 91, 87, 55, 60, 75, 91, 22, 47, 63,
         81, 88, 26, 48, 69, 59, 84, 77, 28, 36, 59, 74, 89, 73, 91, 64, 55, 88, 90, 48,
         73, 97, 98, 40, 93, 50, 78, 60, 44, 77, 82, 51, 53, 65, 98, 59, 94, 91, 52, 44,
         65, 85, 72, 92, 49, 67, 58, 48, 62, 54, 89, 67, 58, 48, 85, 45, 77, 76, 81, 77]

n = len(nilai)

# 3. Cetak data nilai awal (10 data per baris, rata kanan)
print("Data nilai awal :")
k = 0
while k < n:
    print(f"{nilai[k]:3d}", end="")
    if k < n - 1:
        print(", ", end="")
    if (k + 1) % 10 == 0:
        print()
    k += 1
print(garis2)

# 4. Salin data agar data awal tetap utuh
data = nilai[:]

# 5. Proses Bubble Sort
i = 0
while i < n - 1:
    j = 0
    while j < n - i - 1:
        if data[j] > data[j + 1]:
            temp = data[j]
            data[j] = data[j + 1]
            data[j + 1] = temp
        j += 1
    i += 1

# 6. Cetak data terurut (10 data per baris, rata kanan)
print("Data nilai terurut :")
k = 0
while k < n:
    print(f"{data[k]:3d}", end="")
    if k < n - 1:
        print(", ", end="")
    if (k + 1) % 10 == 0:
        print()
    k += 1
print(garis2)
print("selesai".center(50))
print(garis1)
```

6) Print out hasil sc

```
PS C:\Users\user\.vscode> & C:\Users\user\AppData\Local\Programs\Python\Python313\python.exe c:/Users/user/.vscode/SC-MODUL-5-1306625044-DKP.py
==================================================
                PROGRAM SORTING
==================================================
Nama : Dyahayu Khaila Putri
NIM  : 1306625044
--------------------------------------------------
Data nilai awal :
 85, 13, 99, 34, 71, 15, 82, 24, 64, 61,
 67, 99, 50, 68, 25, 37, 32, 27, 14, 91,
 79, 15, 47, 48, 74, 88, 64, 53, 77, 50,
 24, 91, 87, 55, 60, 75, 91, 22, 47, 63,
 81, 88, 26, 48, 69, 59, 84, 77, 28, 36,
 59, 74, 89, 73, 91, 64, 55, 88, 90, 48,
 73, 97, 98, 40, 93, 50, 78, 60, 44, 77,
 82, 51, 53, 65, 98, 59, 94, 91, 52, 44,
 65, 85, 72, 92, 49, 67, 58, 48, 62, 54,
 89, 67, 58, 48, 85, 45, 77, 76, 81, 77
--------------------------------------------------
Data nilai terurut :
 13, 14, 15, 15, 22, 24, 24, 25, 26, 27,
 28, 32, 34, 36, 37, 40, 44, 44, 45, 47,
 47, 48, 48, 48, 48, 48, 49, 50, 50, 50,
 51, 52, 53, 53, 54, 55, 55, 58, 58, 58,
 59, 59, 60, 60, 61, 62, 63, 64, 64, 64,
 65, 65, 67, 67, 67, 68, 69, 71, 72, 73,
 73, 74, 74, 75, 76, 77, 77, 77, 77, 77,
 78, 79, 81, 81, 82, 82, 84, 85, 85, 85,
 87, 88, 88, 88, 89, 89, 90, 91, 91, 91,
 91, 91, 92, 93, 94, 97, 98, 98, 99, 99
--------------------------------------------------
                selesai
==================================================
PS C:\Users\user\.vscode>
```

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